Football Match Prediction

Data Mining Project

HW2

KNTU

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Course: Data Mining

Project Overview

This project focuses on predicting the outcome of football matches using data mining and machine learning techniques. The main goal is to analyze match information and build a predictive system that can classify possible results such as home win, away win, or draw.

Football match prediction is a challenging task because the final result depends on many factors, including team strength, recent performance, match location, competition level, managers, and previous results. By studying these patterns, the project aims to understand which features may influence match outcomes.

Dataset Description

The dataset contains historical football match records, where each row represents one match. The main column is `match_result`, which represents the final match result that should be predicted. The dataset also includes a `match_id` column for identifying each match.

Basic match information is included through columns such as `home_club`, `away_club`, `fixture_date`, `competition`, `competition_code`, and `cup_game`. These columns describe the clubs involved, the date of the match, the competition, and whether the match belongs to a cup tournament.

The dataset also includes manager information using columns such as `home_manager_id` and `away_manager_id`. Manager-related features may help capture differences in team management, strategy, or playing style.

In addition to current match details, the dataset contains historical information about previous matches for both clubs. These recent-history columns describe each club's previous 10 matches, including match dates, whether the club played at home, whether the match was a cup game, goals scored, goals conceded, team rating, opponent rating, manager ID, and competition code.

These historical features are useful because football outcomes are often influenced by recent form, past opponents, competition strength, and match conditions. Therefore, the dataset combines current match information with recent performance indicators to support the prediction task.

Column Description and Explanation

This section explains the meaning of the main columns and the repeated recent-history columns in the dataset.

Main Match Columns

Column Name	Description
`match_id`	A unique identifier for each match record. It helps distinguish one row from another.
`match_result`	The target column of the dataset. This is the final match result that the model should predict, such as home win, away win, or draw.
`home_club`	The name of the club playing as the home team in the current match.
`away_club`	The name of the club playing as the away team in the current match.
`fixture_date`	The date when the match was played or scheduled.
`competition`	The name of the league or competition where the match took place.
`competition_code`	A numerical or encoded identifier for the competition.
`cup_game`	A flag that shows whether the match was part of a cup tournament. Usually, this value is 1 or True for cup games and 0 or False for non-cup games.
`home_manager_id`	An identifier for the manager or coach of the home club.
`away_manager_id`	An identifier for the manager or coach of the away club.

Recent-History Column Structure

Many columns describe the previous matches of the home and away clubs. These columns follow a repeated naming structure:

`club_side_recent_feature_number`

For example:

`home_club_recent_goals_scored_1`

`away_club_recent_team_rating_5`

The meaning of each part is shown below.

Part of Column Name	Meaning
`home_club` or `away_club`	Shows whether the historical information belongs to the current home club or the current away club.
`recent`	Shows that the column describes a previous match, not the current match.

`feature`	Shows the type of information recorded, such as goals scored, goals conceded, rating, manager ID, or competition code.
`number`	Shows which previous match is being described. The number ranges from 1 to 10. Usually, `_1` is the most recent previous match and `_10` is the tenth most recent previous match.

Home Club Recent-History Columns

The following columns describe the previous 10 matches of the current home club.

Column Pattern	**Description**
`home_club_recent_fixture_date_1` to `home_club_recent_fixture_date_10`	The dates of the home club's previous 10 matches.
`home_club_recent_played_at_home_1` to `home_club_recent_played_at_home_10`	Shows whether the home club played each previous match at home.
`home_club_recent_cup_game_1` to `home_club_recent_cup_game_10`	Shows whether each previous match was a cup game.
`home_club_recent_goals_scored_1` to `home_club_recent_goals_scored_10`	The number of goals scored by the home club in each previous match.
`home_club_recent_goals_conceded_1` to `home_club_recent_goals_conceded_10`	The number of goals conceded by the home club in each previous match.
`home_club_recent_team_rating_1` to `home_club_recent_team_rating_10`	The rating or strength score of the home club in each previous match.
`home_club_recent_opponent_rating_1` to `home_club_recent_opponent_rating_10`	The rating or strength score of the opponent faced by the home club in each previous match.
`home_club_recent_manager_id_1` to `home_club_recent_manager_id_10`	The manager or coach ID of the home club during each previous match.
`home_club_recent_competition_code_1` to `home_club_recent_competition_code_10`	The competition code for each previous match played by the home club.

For example, `home_club_recent_goals_scored_1` means the number of goals scored by the current home club in its most recent previous match. Similarly, `home_club_recent_opponent_rating_3` means the rating of the opponent that the current home club faced in its third most recent previous match.

Away Club Recent-History Columns

The following columns describe the previous 10 matches of the current away club.

Column Pattern	**Description**
`away_club_recent_fixture_date_1` to `away_club_recent_fixture_date_10`	The dates of the away club's previous 10 matches.
`away_club_recent_played_at_home_1` to `away_club_recent_played_at_home_10`	Shows whether the away club played each previous match at home.

away_club_recent_cup_game_1 to away_club_recent_cup_game_10	Shows whether each previous match was a cup game.
away_club_recent_goals_scored_1 to away_club_recent_goals_scored_10	The number of goals scored by the away club in each previous match.
away_club_recent_goals_conceded_1 to away_club_recent_goals_conceded_10	The number of goals conceded by the away club in each previous match.
away_club_recent_team_rating_1 to away_club_recent_team_rating_10	The rating or strength score of the away club in each previous match.
away_club_recent_opponent_rating_1 to away_club_recent_opponent_rating_10	The rating or strength score of the opponent faced by the away club in each previous match.
away_club_recent_manager_id_1 to away_club_recent_manager_id_10	The manager or coach ID of the away club during each previous match.
away_club_recent_competition_code_1 to away_club_recent_competition_code_10	The competition code for each previous match played by the away club.

For example, `away_club_recent_goals_conceded_2` means the number of goals conceded by the current away club in its second most recent previous match. Similarly, `away_club_recent_played_at_home_4` shows whether the current away club played its fourth most recent previous match at home.

Meaning of the Numbers 1 to 10

The suffix numbers at the end of the recent-history columns represent the order of previous matches.

Suffix	**Meaning**
_1	The most recent previous match.
_2	The second most recent previous match.
_3	The third most recent previous match.
_4 to _9	Older previous matches in order.
_10	The tenth most recent previous match.

For example, `home_club_recent_goals_scored_1` is more recent than `home_club_recent_goals_scored_10`.

Data Cleaning

Football datasets may contain missing values, duplicated records, inconsistent club names, or columns that are not useful for prediction. These issues should be handled before building the prediction model.

Important cleaning tasks include checking missing values in club, competition, manager, and recent-history columns. Some manager IDs or recent-history values may be missing, especially if previous match information is unavailable. These missing values should be treated carefully so that useful records are not removed unnecessarily.

It is also important to check whether categorical columns such as `home_club`, `away_club`, and `competition` are consistent. Clean and organized data helps the model learn meaningful patterns instead of noise.

Feature Engineering

Feature engineering plays an important role in improving prediction quality. Useful features may be created from the fixture date, club names, home or away status, competition information, manager identifiers, and recent match history.

For example, the `fixture_date` column can be used to extract time-based features such as year, month, or season. The `cup_game` column can help separate cup matches from regular competition matches, since these games may have different patterns.

Recent-history columns can also be used to describe recent club performance. Features based on the last several matches may help represent club form, experience, and consistency. Categorical information such as club names, competition names, and manager IDs can be transformed into numerical form so that it can be used by machine learning algorithms.

Modeling and Evaluation

After preprocessing the data, suitable classification models can be trained to predict football match outcomes. Different approaches may be compared to find the model that performs best on unseen data.

The models can be evaluated using classification metrics such as accuracy, precision, recall, F1-score, or log loss. The final result should provide both match outcome predictions and useful insights into the factors that influence football results.